Assignment - 3

~~Q) The sum of two numbers~~

Q1) A number consists of 2 digits, if 3/5 of 1/5 of the number is 9. Find the sum of its two digits.

→ $\frac{3}{5}$ of $\frac{1}{5}$ → $\frac{3}{5} * \frac{1}{5} x = 9$

$= \frac{3}{25} x = 9 \quad = 3x = 225$

$= x = 225/3 = \underline{75}$

∴ sum of digits $= 7+5 = \underline{12}$

Q2) If the sum and difference of 2 numbers are 20 and 8 respectively. Then the difference of their squares is :

→ Let the numbers be x and y

$x + y = 20 \qquad x - y = 8$

Difference between their squares : $x^2 - y^2 = ?$

$\therefore x^2 - y^2 = (x+y)(x-y)$

$= 20 * 8$

$= \underline{160}$

Q3) The numerator of a certain fraction is 8 less than the denominator. If 3 is added to the numerator and 3 is subtracted from the denominator, the fraction becomes 3/4. Find the original fraction.

→ Original Fraction $= \frac{N}{D} = ?$

Numerator is 8 less than denominator

$D = N + 8$ — ①

Next,

$\frac{N+3}{D-3} = \frac{3}{4} \quad = 4(N+3) = 3(D-3)$

$= 4N + 12 = 3D - 9$

$= 4N - 3D = -9 - 12$

$= 4N - 3D = -21$ — ②

Putting value of D from ① in ②

$4N - 3(N+8) = -21$

$= 4N - 3N - 24 = -21$

$= N - 24 = -21$

$= N = -21 + 24$

$= N = 3$

Putting value in ①, $D = 3 + 8$

$D = 11$

$\therefore$ Original Fraction $= \frac{3}{11}$

(Q4) The sum of four consecutive two digit odd numbers, when divided by 10 becomes a perfect square. Which of the following can possibly one of these four numbers? 21, 25, 41, 67

→ $\underset{n+2}{\underline{n-3, n-1}}, n, \underset{n+2}{\underline{n+1, n+3}}$

$n-3, n-1, n+1, n+3$

Sum $= n-3+n-1+n+1+n+3 = 4n$

$\frac{4n}{10}$ = Perfect square (x) = 1, 4, 9, 16, 25, 36, 49, 64, ...

Assume $x = 4$ i.e $\frac{4n}{10} = 4 = 4n = 40 = n = 10$

$7, 9 \leftarrow n \rightarrow 11, 13$

Assume $x = 16$ i.e $\frac{4n}{10} = 16 = 4n = 160 = n = 40$

$37, 39 \leftarrow n \rightarrow 41, 43$

Assume $x = 36$ i.e $\frac{4n}{10} = 36 = 4n = 360 = n = 90$

$87, 89 \leftarrow n \rightarrow 91, 93$

Q5) In a 4 digit number, the sum of first two digits is equal to that of last two digits. The sum of the first and last digit is equal to the third digit. Finally the sum of second and fourth digits is twice the sum of other two digits. What is the 3rd digit of that number?

→ Let a, b, c, d be the 4 digits

∴ Digit = ~~dcba~~ abcd Require = c

$a+b = c+d$ — ①

$a+d = c$ — ②

$b+d = 2(a+c)$ — ③

Putting value of c from ② in ①

$a+b = a+d+d$

$\cancel{a}+b = \cancel{a}+2d$

$b = 2d$ — ④

Putting ④ and ② in ③

$2d+d = 2(a+a+d)$

$2d+d = 2(2a+2d)$

$3d+\cancel{d} = 4a+\cancel{d}$

$d = 4a$ — ⑤

$a = d/4$

Putting ⑤ in ②

$\frac{d}{4}+d = c$

$\frac{5d}{4} = c$ or $c = \frac{5}{4d}$

As a and d are single digits and $d = 4a$, the satisfying conditions for a is 1, 2.

∴ Value of $d = 4$ or 8

when $d=4$, $\frac{5(4)}{4} = 5$

$d=8$, $\frac{5(8)}{4} = 10$

Since it is single digit, $d=4$ and $c=5$

Q6) Sum of digits of a two-digit number is 10. While when the digits are reversed, the number decreases by 54. Find the changed number.

→ Let the digit be xy – 10 place

So, $x + 10y$

According the statement

$10x + y$

$10x + y = x + 10y - 54$

$10x - x = 10y - y - 54$

$9x = 9y - 54$

$9x - 9y = -54$

$9(x - y) = -54$

Thus, $(x - y) = -6$

$\therefore x - y = -6$ — ①

$x + y = 10$ — ②

Adding ① and ②

$2x = 4$

$x = 2$

Hence, the other digit is 8.

$\therefore$ The number is $\underline{\underline{28}}$

Q7) A Forester wants to plant 44 apple trees, 66 banana trees and 110 mango trees in equal rows (in terms of no. of trees). Also he wants a distinct rows of trees (i.e only one type of tree in a row). The no. of minimum rows required is ..

→ $44 \rightarrow 2 \times 2 \times 11$

$66 \rightarrow 2 \times 3 \times 11$

$110 \rightarrow 2 \times 5 \times 11$

$\therefore$ HCF $(44, 66, 110) \rightarrow 2 \times 11 \rightarrow \underline{22} \rightarrow$ No. of trees in each row

Total trees $= 44 + 66 + 110 = 220$

$\therefore$ Min rows $= \dfrac{\text{Total trees}}{\text{Trees in each row}} = \dfrac{220}{22} = \underline{\underline{10}}$

Q8) X is a five digit number. The digit in ten thousandth place is 1. The number formed by it's digits in units and tens places is divisible by 4. The sum of all digits is divisible by 3. If 5 and 7 also divide X, then X = ?

→ Divisibility of 4 :→ Last 2 digits divisible by 4

Divisibility of 7 :→ Divide the number into groups of 3 digits (from right) and find difference between odd and even places.

Out of all options :- 10080 gets divisible by 7

It also satisfies divisibility of 4. 80/4

Divisibility of 3 → sum of all digits divisible by 3

As last digit is 0 → divisible by 5.

Q9) The difference between a two digit number and a number obtained by interchanging positions of its digits is 36. What is the difference between 2 digits of the number?

→ Let 10's place digit be 'x' and unit place digit be 'y'.

According to question,

$$(10x+y) - (10y+x) = 36$$

$$10x + y - 10y - x = 36$$

$$9x - 9y = 36$$

$$9(x-y) = 36$$

$$x - y = 36/9$$

$$x - y = 4$$

Q10) Find a positive number when increased by 17 is equal to 60 times the reciprocal of the number.

→ According to question, Let the no. be x

$$x+17 = 60 * \frac{1}{x}$$

$$x+17 = \frac{60}{x} = x(x+17) = 60$$

$$= x^2+17x = 60$$

$$= x^2+17x-60 = 0 \quad \text{—} ①$$

$$= x^2-3x+20x-60 = 0$$

$$= x(x-3)+20(x-3) = 0$$

$$= (x+20)(x-3) = 0$$

$$x+20 = 0 \qquad x-3 = 0$$

$$x = -20 \qquad \underline{\underline{x = 3}} \quad \text{(Positive)}$$

Q11) Out of six consecutive natural numbers, if the sum of first three is 27, what is the sum of other three?

→ Let the numbers be; $x, x+1, x+2, x+3, x+4, x+5$

According to question,

$$x+x+1+x+2 = 27$$

$$3x+3 = 27$$

$$3x = 27-3 = 3x = 24$$

$$= \underline{\underline{x = 8}}$$

To find; $(x+3)+(x+4)+(x+5) = ?$

Putting $x=8$

$$(8+3)+(8+4)+(8+5) = ?$$

$$11+12+13 = ? = \underline{\underline{36}}$$

Q12) The sum of three consecutive odd natural numbers each divisible by 3 is 72. What is the largest among them?

→ Assuming the multiples as $3x$, $3(x+1)$, $3(x+2)$

$\therefore 3x + 3x + 3 + 3x + 6 = \cancel{27}\ 72$

$= 9x + 9 = \cancel{27}\ 72$

$= 9x = 72 - 9 = 63$

$\therefore x = 7$

Among, $3x$, $3(x+1)$, $3(x+2)$,

$3(x+2)$ is the largest

Putting $x = 7$ in $3(x+2)$

$= 3(7+2)$

$= 3(9) = \underline{\underline{27}}$

Q13) The sum of squares of 3 consecutive positive numbers is 365. The sum of the numbers is

→ let the consecutive numbers be $(x-1)$, x, $(x+1)$

According to the question,

$(x-1)^2 + x^2 + (x+1)^2 = 365$

$x^2 - 2x + 1 + x^2 + x^2 + 2x + 1 = 365$

$3x^2 + 2 = 365$

$3x^2 = 365 - 2 = 363$

$\therefore x^2 = 363/3 = 121$

$\therefore x = 11$

So, $(x-1) = 10$, $x = 11$, $(x+1) = 12$

$\therefore 10 + 11 + 12 = \underline{\underline{33}}$

Q14) By interchanging the digits of a two digit number, we get a number which is 4 times the original number minus 24. If the unit digit of the original number exceeds it's 10's digit by 7, then original number is :

→ Let the digits be xy

$10x + y$ — ①

∴ Interchanged number, $10y + x$ — ②

According to question

$y = x + 7$ — ③

Putting value of y in ①

$10x + x + 7$

$= 11x + 7$

(Original)

Putting value of y in ②

$10(x+7) + x$

$= 10x + 70 + x$

$= 11x + 70$

(Interchanged)

$11x + 70 = 4(11x + 7) - 24$

$11x + 70 = 44x + 28 - 24$

$70 - 28 + 24 = 44x - 11x$

$66 = 33x$

$\therefore x = 2$

Putting $x = 2$ in ③

$\therefore y = 2 + 7$

$y = 9$

∴ The digit is $\underline{\underline{29}}$

Q15) A man spends $1/3^{rd}$ of his income on food, $2/5^{th}$ of his income on house rent and $1/5^{th}$ of his income on clothes. If he still has Rs. 400 left with him, his income is:

→ Let the total income be Rs. x

$$\therefore x = 1/3\,x + 2/5\,x + 1/5\,x + 400$$

$$x = \frac{x}{3} + \frac{2x}{5} + \frac{x}{5} + 400$$

$$x = \frac{5x}{15} + \frac{6x}{15} + \frac{3x}{15} + 400$$

$$x = \frac{14x}{15} + 400$$

$$x - \frac{14x}{15} = 400$$

$$\frac{15x}{15} - \frac{14x}{15} = 400$$

$$\frac{x}{15} = 400$$

$$\therefore x = 400 \times 15$$

$$x = \underline{\underline{6000}}$$

Q16) The denominator of fraction is 3 more than its numerator, if the numerator is increased by 7 and denominator is decreased by 2, we obtain 2. The sum of numerator and denominator of a fraction is:

→ Let the numerator be 'n', Thus, denominator is $(n+3)$

According to question,

$$\frac{n+7}{(n+3)-2} = 2$$

$$= \frac{n+7}{n+3-2} = 2 \quad = \frac{n+7}{n+1} = 2$$

$$= n+7 = 2(n+1)$$

$$= n+7 = 2n+2$$

$$= 7-2 = 2n-n$$

$$= \underline{\underline{5 = n}}$$

$\therefore n = 5$

Fraction $= \frac{5}{5+3} = \frac{5}{8}$

Sum of numerator and denominator

$= 5+8 = \underline{13}$

Q17) A number exceeds it's one-fifth by 20. The number is :

→ let the number be x

According to question, $x + 20 = \frac{1}{5}x$

$$x + 20 = \frac{x}{5}$$

$$= \frac{x}{5} - \frac{x}{1} = 20$$

$$= \frac{5x - x}{5} = 20$$

$$= \frac{4x}{5} = 20 = 4x = 100 \therefore \underline{\underline{x = 25}}$$

Q18) Express 45 minutes as a fraction of one day.

→ 45 minutes $= \frac{3}{4}$ hours

A day has 24 hours $\therefore$ 3/4 hours of 24 hours

$$\therefore \frac{\frac{3}{4}}{24} = \frac{3}{4} \times \frac{1}{24}$$

$$= \frac{3}{96} = \underline{\underline{\frac{1}{32}}}$$

Q19) 5, 16, 49, 104, ?

→ $a_1 = a = \underline{\underline{5}}$

$a_2 = (11 \times 1) + a_1$

$= (11) + 5 = \underline{\underline{16}}$

$a_3 = (11 \times 3) + a_2$

$= (33) + 16 = \underline{\underline{49}}$

$a_4 = (11 \times 5) + a_3$

$= (55) + 49 = \underline{\underline{104}}$

$\therefore a_5 = (11 \times 7) + a_4$

$= (77) + 104$

$= \underline{\underline{181}}$

Q20) 8, 7, 11, 12, 14, 17, 17, 22, ?

→ Breaking series in alternate numbers

(8, 11, 14, 17, ?) , (7, 12, 17, 22)

↓ ↓

Increasing by 3 Increasing by 5

∴ 17+3 = 20